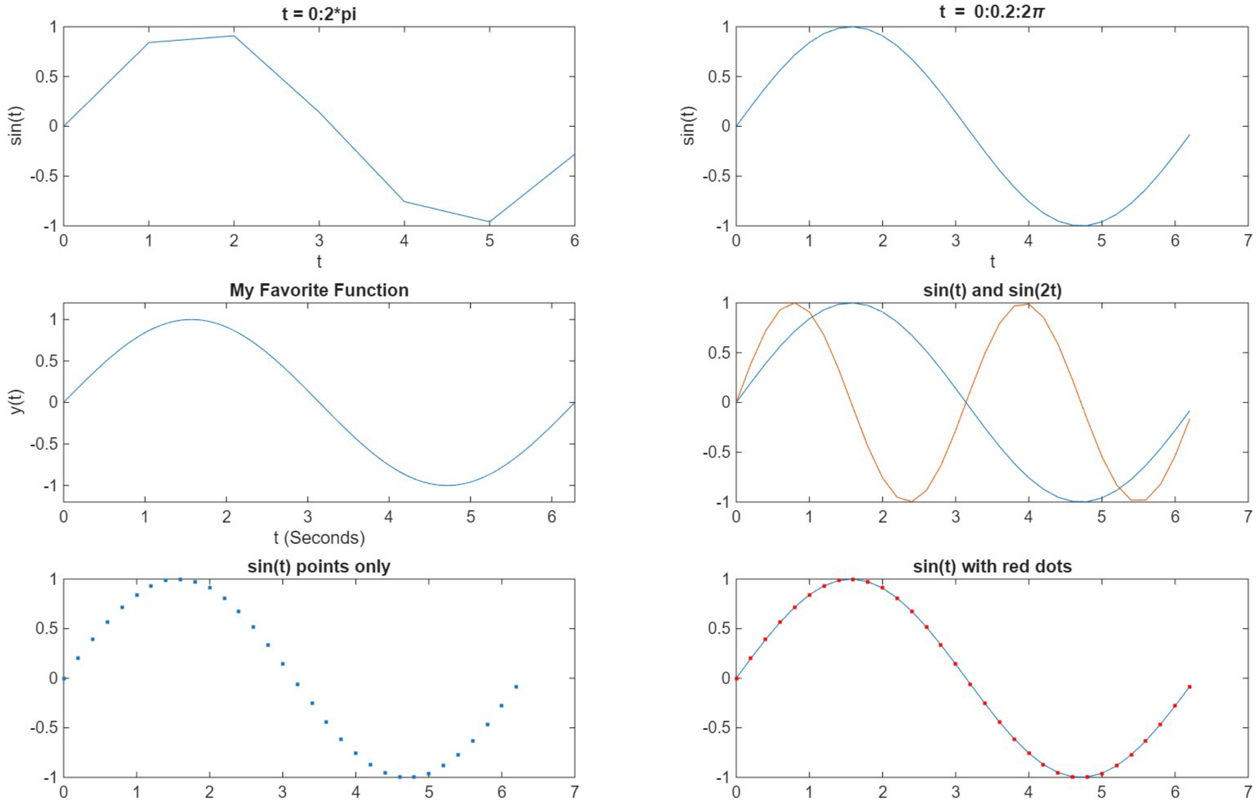
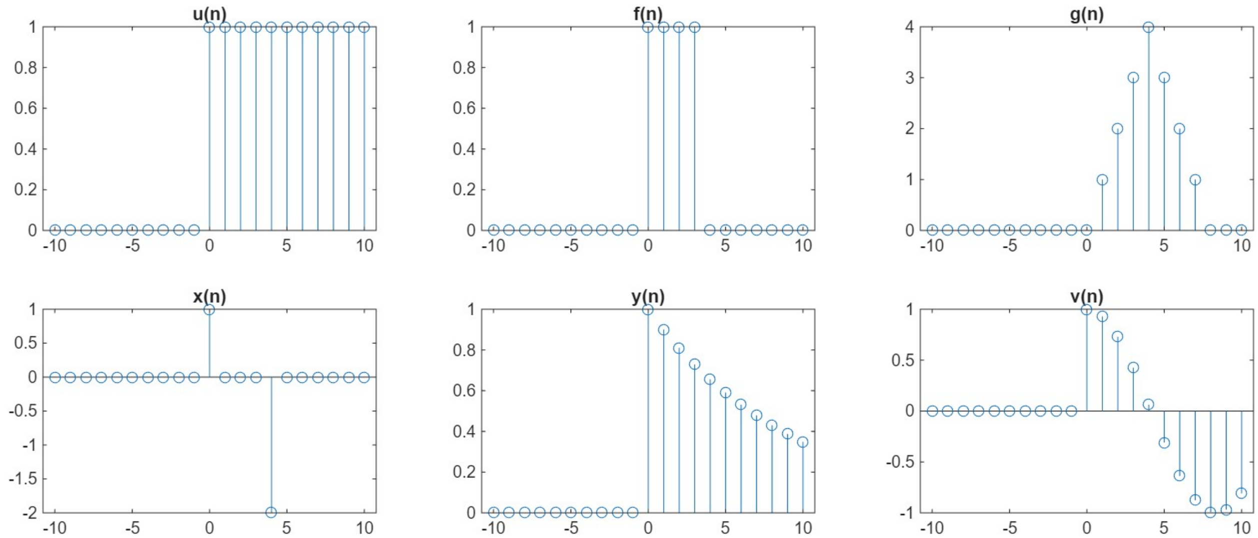
Task 1: Continuous-Time Signals

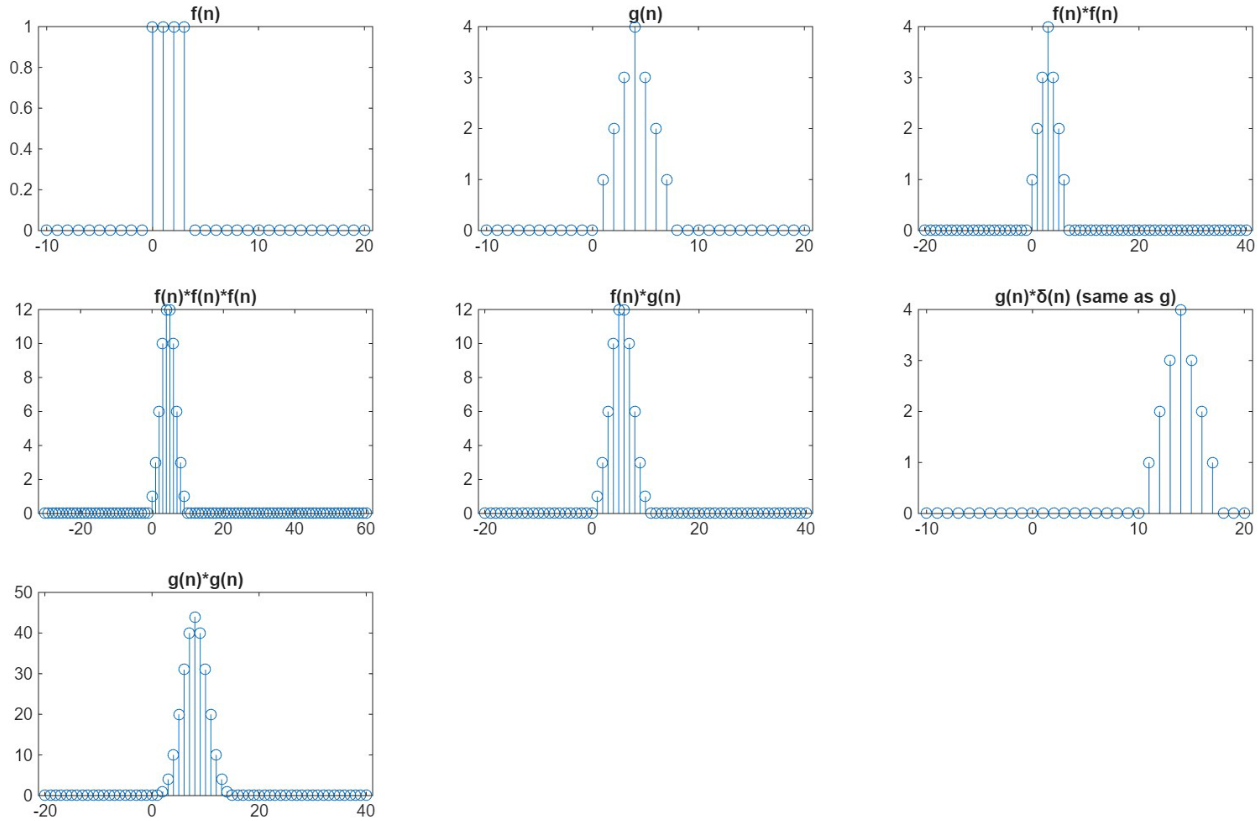


- **Figure: Task 1 – Subplot (3,2,1) – t = 0:2*pi**
  - Only 5 points are plotted due to the large step size, so a smooth curve isn't obtained.
- **Figure: Task 1 – Subplot (3,2,2) – t = 0:0.2:2*pi**
  - Curve looks smoother because more samples are plotted due to the step size of 0.2.
- **Figure: Task 1 – Subplot (3,2,3) – t = 0:0.02:2*pi**
  - With smaller steps, the plot looks more continuous.
- **Figure: Task 1 – Subplot (3,2,6) – sin(t) with red dots**
  - The 'r' makes the points red — 'r' is a color formatting argument.

Task 2: Plotting Discrete Time Signals



Task 3: Convolution



- **Figure: Task 3 – Subplot (3,3,1) – f(n)**
  - *f(n)* is a rectangular pulse (width 4).
- **Figure: Task 3 – Subplot (3,3,2) – g(n)**
  - *g(n)* has a similar shape to $f(n) * f(n)$.
- **Figure: Task 3 – Subplot (3,3,3) – f ∗ f**
  - $f * f$ results in a triangular pulse, wider than *f*.
- **Figure: Task 3 – Subplot (3,3,4) – f ∗ f ∗ f**
  - $f * f * f$ is even wider and smoother.
- **Figure: Task 3 – Subplots (3,3,3) & (3,3,4)**
  - Repeated convolution seems to spread and smooth the signal.
- **Figure: Task 3 – Subplot (3,3,2) vs. Subplot (3,3,3)**
  - *g(n)* looks like $f * f$, which explains their close relationship.

Task 4: Plotting Sampled Signal